AIGC Detection — Accuracy Report

Direction 1: AI-Generated Text Feature Mining & Detection

Dataset: Human ChatGPT Comparison Corpus (HC3)

1. Pipeline Overview

This project implements a multi-dimensional linguistic feature engineering pipeline combined with interpretable machine learning models to distinguish human-written text from AI-generated (ChatGPT) text. The core philosophy is: NOT blindly calling complex deep learning black-box models, but deeply mining the statistical features and linguistic patterns that separate human and AI writing, building a highly interpretable classifier.

Pipeline: Data Loading → Feature Extraction (6 categories, 46-51 features) → Correlation Analysis (|r|>0.85 filtering) → Random Forest / XGBoost / LightGBM → SHAP Attribution Analysis → Multi-Model Feature Importance Consensus.

2. Feature Engineering (6 Categories)

Vocabulary Richness:	TTR, Hapax Legomena Ratio, Honore's R, Vocabulary Density
Readability:	Flesch Reading Ease, Flesch-Kincaid Grade, Avg Word Length, Avg Syllables/Word (EN); Avg Word/Chars, Avg Sent Length (ZH)
POS & Syntax:	Noun/Verb/Adjective/Adverb/Conjunction/Determiner/Punctuation/Pronoun ratios
Dependency Depth:	Max/Avg/Std dependency tree depth, Deep clause ratio (>5), Avg branching factor (EN only, via spaCy)
Transition Words:	7 sub-categories: Sequential, Conclusive, Contrastive, Additive, Causal, Exemplifying, Hedging + Structured marker ratio
Sentiment & Emotion:	VADER compound/neutral/neg/pos, Emotion word ratio, Per-sentence sentiment fluctuation/range/flip ratio ("machine flavor" detection)

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3. HC3 English Results (10,000 samples)

Original features: 51. After correlation filtering (|r|>0.85): 42 features.

Removed 9 redundant features: sentence_count, ttr, avg_syllables_per_word, flesch_kincaid_grade, avg_sent_len, sentiment_fluctuation, trans_sequential, text_len_words, avg_para_len.

3.1 Model Performance

Model	Accuracy	Precision	Recall	F1 Score	AUC
Random Forest	93.00%	0.9479	0.9100	0.9286	0.9798
XGBoost (BEST)	95.10%	0.9583	0.9430	0.9506	0.9898
LightGBM	94.90%	0.9554	0.9420	0.9486	0.9887

5-Fold Cross-Validation (LightGBM): Mean F1 = 0.9394 (+/- 0.0040)

Best Model: XGBoost (F1 = 0.9506, AUC = 0.9898)

3.2 SHAP Top 10 — What Exposes AI Identity?

Rank	Feature	SHAP	Direction	Interpretation
1	unique_bigram_ratio	1.4193	↑ in AI	AI uses more repetitive bigram patterns
2	dep_depth_avg	0.7910	↑ in AI	AI produces deeper clause nesting
3	hapax_ratio	0.6410	↓ in AI	AI uses fewer rare words (less diverse)
4	adv_ratio	0.6175	↓ in AI	Humans use more adverbs (colloquial)
5	vocab_density	0.5275	↓ in AI	AI has lower vocabulary density
6	text_len_chars	0.5098	↓ in AI	Human texts are longer in chars
7	stopword_ratio	0.4575	↓ in AI	Humans use more stopwords
8	sent_len_cv	0.4020	↓ in AI	Human sentence length varies more
9	punct_ratio	0.3110	↓ in AI	Humans use more punctuation variety
10	paragraph_count	0.2871	↑ in AI	AI structures text into more paragraphs

3.3 Multi-Model Consensus (all 3 models agree)

Feature	RF Rank	XGB Rank	LGB Rank	Consensus
unique_bigram_ratio	1	1	1	★★★
dep_depth_avg	3	2	4	★★★
text_len_chars	5	6	3	★★★
hapax_ratio	2	3	9	★★★
vocab_density	7	8	6	★★
sent_len_cv	10	9	2	★★
adv_ratio	8	12	11	★★

3.4 Feature Group Importance

Feature Group	Total SHAP	% of Total
Text Statistics	2.8886	26.6%

POS & Syntax	1.9693	18.1%
Dependency Depth	1.4499	13.3%
Vocabulary Richness	1.2282	11.3%
Sentence Structure	1.0698	9.8%
Transition Words	0.9558	8.8%
Sentiment & Emotion	0.7488	6.9%
Readability	0.5613	5.2%

4. HC3 Chinese Results (10,000 samples)

Original features: 46. After correlation filtering (|r|>0.85): 40 features.

Removed 6 redundant features: avg_sent_len, sentiment_fluctuation, trans_sequential, text_len_words, avg_sent_len_words, ttr_word.

4.1 Model Performance

Model	Accuracy	Precision	Recall	F1 Score	AUC
Random Forest	90.00%	0.9049	0.8940	0.8994	0.9665
XGBoost	92.00%	0.9242	0.9150	0.9196	0.9782
LightGBM (BEST)	92.05%	0.9269	0.9130	0.9199	0.9779

5-Fold Cross-Validation (LightGBM): Mean F1 = 0.9165 (+/- 0.0056)

Best Model: LightGBM (F1 = 0.9199, AUC = 0.9779)

4.2 SHAP Top 10

Rank	Feature	SHAP	Direction	Interpretation
1	unique_bigram_ratio	0.7501	↑ in AI	AI uses more repetitive bigram patterns
2	vocab_density	0.6175	↓ in AI	AI has lower vocabulary density
3	adv_ratio_zh	0.6094	↓ in AI	Humans use more adverbs (colloquial)
4	pronoun_ratio_zh	0.5669	↑ in AI	AI uses more pronouns (formal structure)
5	sent_len_cv	0.5117	↓ in AI	Human sentence length varies more
6	conj_ratio_zh	0.4234	↑ in AI	AI uses more conjunctions (logical flow)
7	honore_r	0.4022	↓ in AI	AI vocabulary richness score lower
8	trans_hedging	0.3764	↑ in AI	AI uses more hedging expressions
9	avg_para_len	0.3322	↑ in AI	AI produces longer paragraphs
10	avg_word_len_chars	0.3258	↓ in AI	Human word length slightly higher

4.3 Multi-Model Consensus

Feature	RF Rank	XGB Rank	LGB Rank	Consensus
unique_bigram_ratio	1	1	15	★★★
adv_ratio_zh	5	11	5	★★
sent_len_cv	2	6	16	★★
vocab_density	7	13	4	★★
honore_r	4	4	17	★★
conj_ratio_zh	9	17	1	★★
pronoun_ratio_zh	12	14	2	★★

4.4 Feature Group Importance

Feature Group	Total SHAP	% of Total
POS & Syntax	2.4501	26.7%

Text Statistics	1.7538	19.1%
Sentence Structure	1.6558	18.0%
Vocabulary Richness	1.4615	15.9%
Transition Words	0.7821	8.5%
Readability (Chinese)	0.6251	6.8%
Sentiment & Emotion	0.4587	5.0%

5. Cross-Language Summary

Language	Best Model	Accuracy	F1 Score	AUC	CV Mean F1
English	XGBoost	95.10%	0.9506	0.9898	0.9394
Chinese	LightGBM	92.05%	0.9199	0.9779	0.9165

6. Key Findings

1. The single most powerful feature across both languages is unique_bigram_ratio (higher in AI text), with top consensus across all 3 models. This indicates AI generates text with more repetitive and formulaic word-pair patterns.

2. Dependency parse tree depth (dep_depth_avg) is the #2 feature for English, confirming that AI text exhibits significantly deeper clause nesting — ChatGPT constructs more hierarchically complex sentences than humans.

3. For Chinese, POS & Syntax features dominate (26.7% of total SHAP), with adverb ratio (lower in AI) and conjunction ratio (higher in AI) being particularly diagnostic. This reflects the "machine flavor" of AI Chinese text.

4. Per-sentence sentiment fluctuation features were removed by correlation filtering in both languages — they correlate with other sentiment features. However, sentiment_neutrality (higher in AI) remains a valid signal.

5. Transition word features (especially hedging and additive categories) contribute ~8-9% of total SHAP importance, confirming that AI overuses structured discourse markers. In Chinese, trans_hedging ranks #8 overall.

6. The XGBoost/LightGBM models achieve 92-95% accuracy using ONLY hand-crafted linguistic features — rivaling deep learning approaches like RoBERTa (HC3 paper reported ~98.8% F1) while providing full interpretability via SHAP.